FOODVERSE AI

Problem Statement

Current food management systems rely on tedious, error-prone manual data entry. Fitness apps require manual calorie logging, commercial kitchens lack automated safety monitoring, and millions of tons of produce are wasted annually globally due to poor spoilage prediction. There is a critical absence of an intelligent system capable of auditing food and hygiene visually.

The Solution

FoodVerse AI resolves this through a fully automated, multimodal intelligence platform. By integrating computer vision, video analysis, and Large Language Model (LLM) reasoning, the system instantly identifies food for automated macro logging, assesses produce degradation to prevent waste, audits hygiene compliance, and generates personalized meal plans from simple inputs.

Project Information

Live Project Demo: https://food-verse-code.vercel.app

Project Codebase: https://github.com/AditiDabral/FoodVerse-Code

Lead Developer: Aditi Dabral

⚡ Core Modules & Features

🍽️ Food Recognition & Nutrition: Automatically identifies dishes from uploaded images and generates estimated macros (calories, protein, carbs, fats) and a health score.

🍓 Freshness Detection: Analyzes visual characteristics (color, texture, discoloration) to classify produce and food as Fresh, Moderately Fresh, or Spoiled.

Restaurant Hygiene Monitor: Evaluates kitchen environments (glove usage, hair-cap compliance, surface cleanliness) to ensure sanitary compliance. Smart AI Chef: Generates custom recipes based on pantry images, voice inputs, and text prompts with step-by-step instructions.

📅 Personalized Meal Planner: Adapts to individual user profiles, biometric parameters, dietary goals, and allergies to create customized multi-day meal plans.

🛠️ Technology Stack

The platform is built using a modern, decoupled architecture designed for highperformance AI inference and rapid UI rendering.

Frontend Layer:

ReactJS (Vite) **Tailwind CSS** **ReactJS (Vite)**

Backend & AI Layer:

FastAPI (Python) **OpenCV** **python** **Groq API**

meta llama model **whisper audio**